O-P-001074
10 September 1965

INTERIM FEDERAL SPECIFICATION

PAD, HOT-COLD TREATMENT, CIRCULATING

This interim Federal specification was developed by the Defense Personnel Support Center, Directorate of Medical Materiel, 2800 South 20th Street, Philadelphia, Pennsylvania, 19101, based upon currently available technical information. It is recommended that Federal agencies use it in Procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 This specification covers one type of controlled temperature pad, suitable for use in therapeutic treatment involving hot or cold applications, both wet and dry.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATION

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585 Boxes, Wood, Wirebound
PPP-B-601 Boxes, Wood, Cleated-Plywood
PPP-B-621 Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 Box, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

FSC 6530

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(Single copies of this specification and other specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-Vaporproof, Flexible

Military Standards:

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Standards Association, Incorporated

ASA Standard

ASA C73.11 - 1963	American Standard Outlet Attachment Plug Caps and Appliance Plugs.

(Copies of ASA Standards may be obtained upon application to the American Standards Association, Incorporated, 10 East 40th Street, New York 16, N.Y.)

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3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit to the contracting agency prior to delivery proof that the equipment he will supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. (UL499) The label or listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the equipment conforms to this requirement.

3.1.2 In lieu of the label or listing by the Underwriters' Laboratories Inc., the contractor may submit prior to delivery independent proof, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, and satisfactory to the contracting agency, that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.1.3 Compliance with Underwriters' Laboratories, Inc., requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material. Material used in the fabrication of the pad shall be as specified herein, of the highest quality throughout, and suitable in all respects for the purpose intended.

3.3 Construction. Pad shall consist of the following:

(a) Sealed electric motor and pump housing provided with a 5-foot minimum length of cord. The line shall be furnished with a plug for use with receptacles, specified in ASA C73.11 - 1963 for Caps and Receptacles, 125 volts, 15 amperes, 2 poles, 3 wire, grounding type. All electrical components shall conform to the requirements of the National Electrical Code.

(b) Water reservoir.
(c) Heat exchanger.
(d) Adjustable thermostat with two keys.
(e) Protector thermostat.
(f) Pilot light.
(g) Two pairs of quick disconnect valves.
(h) Two tubing connectors, 1-$\frac{1}{2}$ inches $\pm$ 1/8 inch long.
(i) Flexible vinyl plastic pad, 12 inches $\pm \frac{1}{2}$ inch wide by 16-3/4 inches $\pm \frac{1}{2}$ inch long, with a pattern of twin passages, together with not less than 10 feet of twin vinyl plastic tubing fitted with disconnect valves to mate with the control unit connectors. The two external tubes shall be preflanged, then heat sealed onto the pad, forming a lap type of seal.

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3.4 Operation. The motor shall be capable of operation on 115 volts, 60 cycles, alternating current. The heating element shall be rated at 200 watts ± 5 percent at 115 volts, alternating current. The motor assembly shall be rated at 23 watts ± 1 watt at 115 volts, alternating current. The heating element shall be capable of maintaining the water temperature to an accuracy of ± 1° Fahrenheit (F) through a range of 85° F. to 115° F., as prescribed. The pumping unit shall also be capable of operation without the heating element. The pad shall operate as specified, when tested as specifie in 4.4.1. Electrical parts shall pass the dielectric test specified in 4.4.2

3.5 Repair parts. Each unit shall be furnished with the following repair parts:

One cup and liner assembly.
One upper reservoir.
One square seal ring gasket.

3.6 Identification marking. Each unit shall have a nameplate securely attached to the motor and pump housing, bearing the following information:

Name or registered trademark of manufacturer
Stock number
Contract or purchase order number
Electrical ratings
U.S.

Marking shall be distinct and legible. Marking secured by adhesives shall not be used.

3.7 Operating instructions. Operating instructions shall be securely affixed to the pump housing.

3.8 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. This information shall specifically include instructions, drawings, and power wiring. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitable indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable

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after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing-in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located.

When two or more containers are required for packing a disassembled item (components of a single stock number), set markings shall be applied in accordance with MIL-STD-129. The envelope shall be affixed to the container designated as No. 1 of the set. The second set and, if necessary, the remainder of the first set of service data, shall be attached to the unit inside the exterior container. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.9 Workmanship. Workmanship shall be first class throughout. The pads shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that the supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examinations (such as visual, auditory, etc, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Heating element	Power consumed	3.4	4.4.1	L8
Motor assembly	Power consumed	3.4	4.4.1	L8
Water	Temperature	3.4	4.4.1	100%
Electrical unit	Dielectric strength	3.4	4.4.2	L8

4.2.3 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The pads shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of Defects - Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Any component part missing, broken, or unusable.
102 Any component not free of cracks, rupture, or tears.
103 Any threaded parts stripped; or not functioning properly.
104 Quick disconnect valve does not operate properly.
105 Any exposed part not free from burs or sharpness.
106 Wire or plug not as specified.
107 Service data missing or not complete.

Minor

201 Identification marking missing, incomplete, illegible or not properly attached.
202 Finish not free of scratches or not uniformly applied.
203 Unit not clean, not free of dirt, oil, or grease.

* Examination shall not be restricted to the characteristics listed above.

4.4 Tests. Tests shall be conducted,as necessary, to determine compliance with specification requirements.

4.4.1 Operations test. The unit, assembled in accordance with the manufacturer's instructions, shall be operated on 115 volts, alternating current, for 1 hour. Data shall be recorded to indicate compliance with requirements contained in 3.4.

4.4.2 Dielectric strength test. While the unit is thoroughly heated, and with all current-carrying parts in the circuit, a test potential of 900 volts shall be developed between the current-carrying parts and one of the water outlet pipes and maintained for 1 minute.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) pad, as specified, constitutes one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each exterior container, as applicable, fc the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit preparation	Intermediate package	Exterior container
1 unit	Not required	2 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit preparation.

5.1.3.1.1 Control unit. The water reservoir shall be protected by means of a corrugated fiberboard sleeve of suitable size and design. Keys shall be packaged in an envelope of suitable size and design. The control unit, with the envelope containing the keys, shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636,- type CF, class domestic. Carton shall be designed with suitable interior fittings, or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Carton shall be adequately secured to prevent accidental opening.

5.1.3.1.2 Pad and tubing. The plastic pad and plastic tubing shall be wound on a paperboard tube of appropriate size, in such a manner that the pad and tubing shall not kink or be permanently deformed . The pad and tubing shall be protected by means of a suitable corrugated sleeve or liner, and packaged in a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Carton shall be adequately secured tp prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size,and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

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5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each carton shall be marked with the stock number, item identification, and name of contractor. In addition, cartons containing the control unit shall be marked "Part 1 of 2", and cartons containing pad and tubing shall be marked "Part 2 of 2".

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5.3.2 Exterior container: Exterior container shall be marked in accordance with Military Standard MIL-STD-129. In addition, the following legend shall appear:

"NOTE: EACH UNIT CONSISTS OF PART 1 AND PART 2".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Section of applicable levels of packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8278

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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PHILADELPHIA, PA. 19101

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER

ATTN: DIRECTORATE MEDICAL MATERIEL

2800 SOUTH 20TH STREET

PHILADELPHIA, PA. 19101

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